

where they capture not only GABA that diffuses away from the synapse, but also neuroactive steroids synthesized and released by glia

- GABA binding at these receptors hypothetically regulates tonic inhibition
- Brexanolone binds to the neuroactive steroid site, acting as a positive allosteric modulator to enhance the effects of GABA binding, i.e., to increase tonic GABAergic tone
- Only known therapeutic agent for depression with positive allosteric modulation of the benzodiazepine-insensitive GABA-A receptor

How Long Until It Works

- Clinical improvement may be evident as early as 12–24 hours into the infusion
- Improvement may continue over the 60-hour infusion regimen

If It Works

- In clinical trials, antidepressant effect was sustained at 30 days post-infusion

If It Doesn't Work

- Consider psychotherapy and/or initiating treatment with an oral antidepressant

Best Augmenting Combos for Partial Response or Treatment Resistance

- Brexanolone is a short-term treatment
- Some patients may require initiation of a longer-term oral antidepressant

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at GABA-A receptors

Notable Side Effects

- Sedation, dizziness, dry mouth, flushing/hot flush

Life-Threatening or Dangerous Side Effects

- Excessive sedation or sudden loss of consciousness
- Syncope, presyncope
- Theoretical activation or emergence of suicidal ideation

Weight Gain



- None

Sedation



- Many experience and/or can be significant in amount
- Excessive sedation with sudden loss of consciousness can occur

What to Do About Side Effects

- If excessive sedation occurs at any time during the infusion, stop the infusion until the symptoms resolve; the infusion may be resumed at the same or lower dose as clinically appropriate
- If hypoxia occurs at any time during the infusion, immediately stop the infusion; after hypoxia, the infusion should not be resumed
- For other side effects, consider a maximum dose of 60 $\hat{I}$ ^{1/4} g/kg/hour (hours 24 to 52)

Best Augmenting Agents for Side Effects

- None

Dosing and Use

Usual Dosage Range

- Administered as a continuous intravenous infusion over 60 hours, titrated from 30 $\hat{1}/4$ g/kg/hour up to 90 $\hat{1}/4$ g/kg/hour and back down to 30 $\hat{1}/4$ g/kg/hour on a set schedule (see How to Dose)

Dosage Forms

- Injection 100 mg/20 mL (5 mg/mL) single-dose vial

How to Dose

- Healthcare provider must be available on site for the duration of the infusion in order to continuously monitor the patient and intervene as necessary
- 0 to 4 hours: initial dose 30 $\hat{1}/4$ g/kg/hour
- 4 to 24 hours: increase dosage to 60 $\hat{1}/4$ g/kg/hour
- 24 to 52 hours: increase dosage to 90 $\hat{1}/4$ g/kg/hour; alternatively consider 60 $\hat{1}/4$ g/kg/hour for those who do not tolerate 90 $\hat{1}/4$ g/kg/hour
- 52 to 56 hours: decrease dosage to 60 $\hat{1}/4$ g/kg/hour
- 56 to 60 hours: decrease dosage to 30 $\hat{1}/4$ g/kg/hour
- Dilution is required prior to administration



Dosing Tips

- Because of the risk of excessive sedation and loss of consciousness, patients must be monitored continuously for the duration of the infusion

- Monitor patients for hypoxia using continuous pulse oximetry equipped with an alarm
- Monitor for excessive sedation every 2 hours during planned, non-sleep periods
- Initiate brexanolone early enough during the day to allow for recognition of excessive sedation
- Patients must be accompanied during interactions with their children
- Brexanolone is supplied as a concentrated solution that must be diluted prior to administration; after dilution the product can be stored in infusion bags under refrigerated conditions for up to 96 hours; however, because the diluted product can be used for only 12 hours at room temperature, each 60-hour infusion will require preparation of at least 5 infusion bags

Overdose

- Limited data

Long-Term Use

- Not intended for long-term use

Habit Forming

- Schedule IV

How to Stop

- Down-titrate per the standard dosing regimen (see How to Dose)

Pharmacokinetics

- Terminal half-life approximately 9 hours
- Extensively metabolized by non-CYP-based pathways; metabolites are inactive



Drug Interactions

- Increased depressive effects when taken with other CNS depressants



Other Warnings/Precautions

- Brexanolone can cause excessive sedation and sudden loss of consciousness; for that reason, brexanolone is available only through a Risk Evaluation and Mitigation Strategy (REMS); the patient must be enrolled in the program prior to administration of brexanolone
- Brexanolone can cause hypoxia; monitor patients using pulse oximetry and stop the infusion if hypoxia occurs

Do Not Use

- If there is a proven allergy to brexanolone

Special Populations

Renal Impairment

- Dose adjustment not necessary for mild, moderate, or severe renal impairment
- Avoid use in patients with end-stage renal disease, due to the potential accumulation of the stabilizing agent, betadex sulfobutyl ether sodium

Hepatic Impairment

- Dose adjustment not necessary

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution

Elderly

- Not studied

Children and Adolescents

- Safety and efficacy have not been established

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy

letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- In animal studies, malformations were not seen in rats or rabbits at levels up to 5 and 6 times the maximum recommended human dose (MRHD), respectively
- Developmental toxicities were seen in rats and rabbits at 5 and >3 times the MRHD, respectively, and reproductive toxicities were seen in rabbits at >3 the MRHD
- When administered during pregnancy and lactation in rats, lower pup survival occurred at doses >2 times the MRHD and a neurobehavioral deficit in female offspring occurred at doses 5 times the MRHD
- In published animal studies, administration of other drugs that enhance GABAergic inhibition to neonatal rats caused widespread apoptotic neurodegeneration during the period of brain development that corresponds to the third trimester of pregnancy in humans
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk

- In a study of 12 healthy adult lactating women treated with intravenous brexanolone according to the recommended 60-hour regimen, concentrations of brexanolone in breast milk were low (<10 ng/mL) in >95% of women; the calculated maximum relative infant dose during infusion was 1–2%

The Art of Psychopharmacology

Potential Advantages

- Rapid response

Potential Disadvantages

- Can only be administered in a Zulresso REMS-certified healthcare setting

Primary Target Symptoms

- Depressed mood

Pearls

- Brexanolone is the first medication approved for the treatment of postpartum depression
- Brexanolone is first-in-class as an antidepressant with a GABAergic mechanism

- A related neuroactive steroid is currently being studied in an oral formulation for the treatment of postpartum depression and unipolar depression

Suggested Reading

Cooper MC , Kilvert HS , Hodgkins P , Roskell NS , Eldar-Lissai A . Using matching-adjusted indirect comparisons and network meta-analyses to compare efficacy of brexanolone injection with selective serotonin reuptake inhibitors for treating postpartum depression . *CNS Drugs* 2019 ;**33** (10):1039–52 .

Meltzer-Brody S , Colquhoun H , Riesenberger R , et al. Brexanolone injection in post-partum depression: two multicentre, double-blind, randomised, placebo-controlled, phase 3 trials . *Lancet* 2018 ;**392** (10152):1058–70 .

Zheng W , Cai DB , Zheng W , et al. Brexanolone for postpartum depression: a meta-analysis of randomized controlled studies . *Psychiatry Res* 2019 ;**279** :83–9 .

Zorumski CF , Paul SM , Covey DF , Mennerick S . Neurosteroids as novel antidepressants and anxiolytics: GABA-A receptors and beyond . *Neurobiol Stress* 2019 ;**11** :100196 .

Brexpiprazole

Therapeutics

Brands

- Rexulti

see index for additional brand names

Generic?

No

Class

- Dopamine partial agonist (dopamine-serotonin partial agonist, dopamine stabilizer, atypical antipsychotic, third-generation antipsychotic; sometimes included as a second-generation antipsychotic; also a potential mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia**
- **Treatment-resistant depression (adjunct)**
- Acute mania/mixed mania
- Other psychotic disorders

- Bipolar maintenance
- Bipolar depression
- Behavioral disturbances in dementia
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- * Partial agonism at dopamine 2 receptors
 - Theoretically reduces dopamine output when dopamine concentrations are high, thus improving positive symptoms and mediating antipsychotic actions
 - Theoretically increases dopamine output when dopamine concentrations are low, thus improving cognitive, negative, and mood symptoms
 - Interactions at a myriad of other neurotransmitter receptors may contribute to brexpiprazole's efficacy
 - Partial agonist at 5HT1A receptors, which may be beneficial for mood, anxiety, and cognition in a number of disorders
 - Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms

- Blockade of alpha 1B receptors may reduce arousal symptoms in posttraumatic stress disorder and in agitation associated with dementia as well as motor side effects such as akathisia
- Blockade of alpha 2C receptors may contribute to antidepressant actions
- Actions at dopamine 3 receptors could theoretically contribute to brexpiprazole's efficacy
- Blocks serotonin 7 receptors, which may be beneficial for mood, cognitive impairment, and negative symptoms in schizophrenia, and also in bipolar disorder and major depressive disorder

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- For psychosis, classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients may require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms
- For depression, onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks

If It Works –for Schizophrenia

- Most often reduces positive symptoms but does not eliminate them

- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment

If It Works –for Depression

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped

- Continue treatment until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work –for Schizophrenia

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection
- Consider initiating rehabilitation and psychotherapy
- Consider presence of concomitant drug abuse

If It Doesn't Work –for Depression

- Some patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder



Best Augmenting Combos for Partial Response or Treatment Resistance

- For depression, brexpiprazole is itself an augmenting agent
- Valproic acid (valproate, divalproex, divalproex ER)
- Mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Lithium
- Topiramate
- Benzodiazepines

Tests

Before starting any atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting any atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when

initiating or switching antipsychotics

* Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight

- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness and clouding of sensorium, even coma

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and brexpiprazole should be discontinued at the first sign of decline in WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Partial agonist actions at dopamine 2 receptors in the striatum can cause akathisia and drug-induced parkinsonism
- Partial agonist actions at dopamine 2 receptors and 5HT1A receptors can also cause nausea, occasional vomiting, and activating side effects
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- Weight gain
- Akathisia (dose-dependent), restlessness (dose-dependent), anxiety
- Sedation, headache
- Tardive dyskinesia (TD) (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Rare impulse control problems

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- As a class, antidepressants have been reported to increase the risk of suicidal thoughts and behaviors in children and young adults
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures



Weight Gain

- Occurs in a significant minority



Sedation

- Occurs in a significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait

- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Schizophrenia: 2–4 mg once daily
- Depression: 2 mg once daily

Dosage Forms

- Tablet 0.25 mg, 0.5 mg, 1 mg, 2 mg, 3 mg, 4 mg

How to Dose

- Schizophrenia: Initial 1 mg once daily for days 1–4; increase to 2 mg once daily for days 5–7; increase to 4 mg once daily on day 8; maximum dose 4 mg once daily
- Depression: initial 0.5–1 mg once daily; increase in weekly intervals up to 1 mg once daily and then up to 2 mg once daily; maximum dose 3 mg once daily



Dosing Tips

- Can be taken with or without food Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine Children and elderly should generally be dosed at the lower end of the dosage spectrum Treatment should be suspended if absolute neutrophil count falls below 1000/mm³

Overdose

- Limited experience

Long-Term Use

- Safety and efficacy demonstrated in schizophrenia in a maintenance study lasting over 1 year
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Because clinical experience is lacking, down-titration may be prudent, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- However, the long half-life suggests that it may be possible to stop brexpiprazole abruptly
- See Switching section of individual agents for how to stop brexpiprazole
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms, but less likely with brexpiprazole due to its long half-life

Pharmacokinetics

- Mean half-life 91 hours (brexpiprazole) and 86 hours (major metabolite DM-3411)

- Metabolized primarily by CYP450 2D6 and CYP450 3A4

Drug Interactions

- In patients receiving a strong/moderate CYP450 3A4 inhibitor (e.g., ketoconazole), brexpiprazole should be administered at half the usual dose
- In patients receiving a strong CYP450 3A4 inducer (e.g., carbamazepine), brexpiprazole should be administered at double the usual dose
- In patients with schizophrenia who are receiving a strong/moderate CYP450 2D6 inhibitor (e.g., quinidine) or who are known CYP450 2D6 poor metabolizers, brexpiprazole should be administered at half the usual dose
- However, clinical trials in major depressive disorder took into account the potential concomitant administration of strong CYP450 2D6 inhibitors (e.g., paroxetine, fluoxetine), so the dose of brexpiprazole does not need to be adjusted in these cases
- In patients receiving both a strong/moderate CYP450 3A4 inhibitor and a strong/moderate CYP450 2D6 inhibitor, brexpiprazole should be administered at one-quarter the usual dose
- In patients receiving a strong/moderate CYP450 3A4 inhibitor who are known CYP450 2D6 poor metabolizers, brexpiprazole should be administered at one-quarter the usual dose
- May increase effects of antihypertensive agents

- May antagonize levodopa, dopamine agonists

Other Warnings/Precautions

- There have been reports of problems with impulse control in patients taking aripiprazole, including compulsive gambling, shopping, binge eating, and sexual activity; use caution when prescribing to patients at high risk for impulse-control problems (e.g., patients with bipolar disorder, impulsive personality, obsessive-compulsive disorder, substance use disorders) and monitor all patients for emergence of these symptoms; dose should be lowered or discontinued if impulse-control problems manifest
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and brexpiprazole should be used cautiously in patients at risk for aspiration pneumonia